

METIMMOX-1 Model Input Parameters

Base Case, Deterministic Sensitivity Analysis (DSA), and Probabilistic Sensitivity Analysis (PSA)
Parameters

Ben Geisler

2025-10-13

Table of contents

1	Summart	1
2	Global Model Parameters	1
3	Model Input Parameters	2
4	Technical Notes	2
4.1	Model Structure	2
4.2	Uncertainty Analysis Methods	2
4.3	Data Sources and Assumptions	3

1 Summart

This report provides an overview of all input parameters used in the METIMMOX-1 health economic model, organized into global model settings and variable parameters across base case, DSA, and PSA analyses.

2 Global Model Parameters

These parameters define the overall model structure and are held constant across all analyses.

Table 1: Global Model Parameters

Parameter	Value	Description	Source
Time Horizon	520 weeks (10 years)	Model time horizon	Clinical follow-up period
Cycle Length	1 week	Cycle length for discrete-time model	Model assumption
Discount Rate	4% per annum	Annual discount rate for costs and effects	Model specification
WTP Threshold	€51,000 per QALY	Willingness-to-pay threshold	Norwegian threshold based on severity (14.5 QALYs lost for diagnosis, based on a health 65-year old)
PSA Sample Size	5,000 iterations	Number of Monte Carlo iterations for PSA	Computational adequacy
Bootstrap Samples	5,000 samples	Bootstrap samples for survival uncertainty	Statistical adequacy
DSA Variation Range	±20%	Parameter variation range for DSA	Standard practice

3 Model Input Parameters

This table presents all variable model parameters organized by type, showing their base case values, DSA variation ranges, and PSA distribution specifications.

Table 2: Model Input Parameters by Type

Type	Parameter	Base Case	DSA Range	PSA Distribution
Cost Parameters				
Cost	Nivolumab per administration	€13,923	€11,138 - €16,708	Gamma (shape=25, rate=0.0018)
Cost	FLOX per administration	€427	€342 - €512	Gamma (shape=25, rate=0.0586)
Cost	CT scan	€386	€309 - €463	Gamma (shape=25, rate=0.0648)
Cost	Blood tests	€0	Not varied	Fixed at 0
Cost	NGS testing	€1,439	€1,151 - €1,727	Gamma (shape=25, rate=0.0174)
Cost	Standard outpatient visit	€33	Not varied	Gamma (shape=25, rate=0.7576)
Cost	Baseline visit	€530	Not varied	Gamma (shape=25, rate=0.0472)
Cost	Follow-up visit	€33	Not varied	Gamma (shape=25, rate=0.7576)
Cost	End-of-life care	€13,803	€11,042 - €16,564	Gamma (shape=25, rate=0.0018)
Utility Parameters				
Utility	Non-progressed state	0.9077	0.7262 - 1.0000	Beta (=41.0, =4.2)
Utility	Progressed state	0.9005	0.7204 - 1.0000	Beta (=40.6, =4.5)
Prevalence Parameters				
Prevalence	CRP biomarker positive	30%	Not varied	Fixed
Prevalence	TLR biomarker positive	25%	Not varied	Fixed
Prevalence	TMB/BRAF biomarker positive	20%	Not varied	Fixed
Survival Parameters				
Survival	OS and PFS curves by biomarker status	Fitted parametric survival models	Not varied	Bootstrap sampling (5,000 samples)

Notes:

DSA ranges calculated as base case $\pm$ 20%. Utility upper bounds capped at 1.0.

PSA: Gamma distributions for costs (CV=20%), Beta for utilities (CV=15%).

Survival curves use bootstrap sampling with age and sex adjustments.

4 Technical Notes

4.1 Model Structure

- **Time horizon:** 520 weeks (10 years) based on long-term follow-up requirements
- **Cycle length:** 1 week for precise timing of treatments and assessments

- **Perspective:** Healthcare system costs and patient-reported health outcomes
- **Population:** Metastatic colorectal cancer patients eligible for biomarker testing

4.2 Uncertainty Analysis Methods

- **DSA:** Parameters varied individually by $\pm$ 20% while others held at base case
- **PSA:** Monte Carlo simulation with 5,000 iterations using appropriate distributions
- **Survival uncertainty:** Bootstrap resampling (5,000 samples) preserves correlation structure
- **Biomarker strategies:** Population-weighted averages account for biomarker prevalence

4.3 Data Sources and Assumptions

- **Clinical effectiveness:** METIMMOX-1 randomized controlled trial data
- **Costs:** National healthcare tariffs, drug pricing databases, and published literature
- **Utilities:** EQ-5D-5L patient-reported outcomes from trial participants
- **Treatment schedules:** Protocol-specified dosing and monitoring requirements

Report completed on: 2025-10-13

Repository: ben-geisler/METIMMOX-1

Report version: 1.0